

Bayes' Theorem

Remember $P(A|B) = \frac{P(A \cap B)}{P(B)}$

1) The law of total probabilities

This law is found as the denominator of the Bayes' theorem formula. On the data sheet, this is shown as the following:

$$\sum_{i=1}^n P(A_i) \cdot P(B|A_i)$$

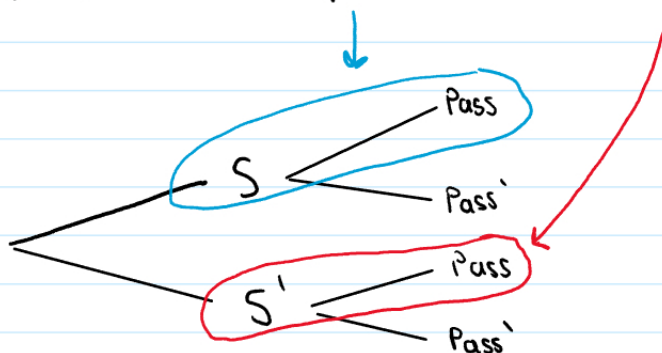
In context, say the probability you pass an exam is denoted as $P(\text{pass})$ where you either study, $P(S)$, or don't study, $P(S')$.

To find the total probability of passing, you calculate the sum of the chance of studying and passing & the chance of not studying and passing.

Using the law of total probabilities, this can be simplified into the following:

$$P(\text{pass}) = P(S) \cdot P(\text{pass}|S) + P(S') \cdot P(\text{pass}|S')$$

Since:



We don't use $P(\text{pass}) \cdot P(S)$ as it assumes events are independent.

Instead, we know to use this law since rearranging the following gives our law.

$$P(\text{pass}|S) = \frac{P(\text{pass} \cap S)}{P(S)}$$

↓

$$P(\text{pass} \cap S) = P(S) \cdot P(\text{pass}|S)$$

This formula doesn't assume independence of events, thus it can be used everywhere. However, it does assume that each event is mutually exclusive and the list is exhaustive.

↑ e.g. $P((\text{pass}|S) \cap (\text{pass}|S')) = 0$

↑ Includes everything in relation to it. e.g. not only $P(\text{pass}|S)$ but $P(\text{pass}|S)$ & $P(\text{pass}|S')$

2) Understanding the formula

The Bayes' theorem formula can be simplified into this: "Given that outcome B happened, what's the chance it was caused by A?"

$$P(A|B) = \frac{P(A) \cdot P(B|A)}{P(B)}$$

OR

$$P(A|B) = \frac{P(A) \cdot P(B|A)}{\sum_{i=1}^n P(A_i) \cdot P(B|A_i)}$$

Where $P(A|B)$ is the Posterior : What we're trying to find

- $P(A)$ is the prior : what we knew before the new evidence
- $P(B|A)$ is the likelihood : If A is true, how likely is B
- $P(B)$ is the total probability of all evidence occurring.
 - > The Sum of all paths.

Simply, if you hear a "meow" (B) and you want to find if it came from a Cat (A), then the "meow" is your evidence and the cat is the prior knowledge.

The top of the division will be the chance a cat exists and makes that sound, with the bottom of the division being the total probability you heard a "meow" from anything.

e.g.

Scenario

Suppose you are listening to sound recordings from a suburban garden:

- 10% of the recorded animals are actually Cats (A), meaning $P(A) = 0.10$. PNG
- 90% of the recorded animals are Other Animals (A'). PNG

Now consider the sound evidence, a Meow (B): PNG

- If the animal is a Cat (A), the probability it makes a meow sound is 0.80 ($P(B|A) = 0.80$). PNG
- If the animal is Not a Cat (A') (e.g., a parrot or a fox), the probability it accidentally makes a sound mistakable for a meow is 0.05 ($P(B|A') = 0.05$). PNG

Your Task

You hear a meow sound (B). Calculate the posterior probability that the animal is actually a Cat, $P(A|B)$. PNG

$$P(A|B) = \frac{P(A) \cdot P(B|A)}{\sum P(A_i) P(B|A_i)} = \frac{0.10 \cdot 0.80}{(0.10 \cdot 0.80) + (0.90 \cdot 0.05)} = 0.640 \checkmark$$

↑
or in other words
 $P(B)$

↑
chance of a
meow and it
being a cat

↑
chance of a meow but it
not being a cat